# Mrinal Pandey

Mp755@cornell.edu ❖ (607) 592-8172 ❖ Ithaca, NY

**LinkedIn:** www.linkedin.com/in/mrinal-pandey

**Google Scholar:** https://scholar.google.com/citations?user=B7w0ifAAAAAJ&hl=en&oi=ao

## PROFESSIONAL SUMMARY

Ph.D. candidate in Biological Engineering at Cornell University with 7+ years of oncology-focused research experience. Expert in **tumor microenvironment, mechanotransduction, and translational oncology models**, integrating **microfluidics, 3D tumor spheroids, RNA sequencing, and imaging** to uncover mechanisms of cancer progression. Skilled in cross-disciplinary collaboration, experimental design, and dissemination of high-impact findings. Dedicated to advancing oncology drug discovery through rigorous science, innovation, and teamwork.

## EDUCATION

**Cornell University** — Ph.D. Candidate, Biological Engineering (GPA 3.9/4.0)
Jan 2020 – Present, Ithaca, NY, USA
• Member of Alpha Epsilon – Biological Engineering Honors Society

**KTH Royal Institute of Technology** — M.S., Medical Biotechnology (GPA 4.5/5)
Aug 2017 – Oct 2019, Stockholm, Sweden

## RESEARCH EXPERIENCE

**Cornell University — Ph.D. Candidate & Research Assistant**
Jan 2020 – Present, Ithaca, NY
• Led research on **mechanical regulation of tumor invasion and immune evasion** in 3D breast cancer organoid models.

• Applied a custom microfluidic compression platform to quantify viscoelastic properties of 3D breast tumor organoids and investigate how mechanical forces influence cellular signaling.

• Integrated **bulk RNA sequencing, NK cell co-culture assays, and quantitative immunofluorescence** to dissect compression-mediated regulation of **STAT1 signaling and immune evasion mechanisms** in breast cancer.

• Generated insights into **ECM remodeling, mechanotransduction, and tumor–immune interactions** that align with clinical drivers of progression and therapeutic resistance.

**SciLifeLab — Master's Thesis Researcher**
Sept 2018 – June 2019, Stockholm, Sweden
• Developed and validated immunoassays for host cell protein detection using microfluidics, ELISA, and mass spectrometry.
• Enhanced assay sensitivity and specificity — experience relevant to biomarker assay development.

## CORE COMPETENCIES

**Oncology & Translational Biology**
- Tumor microenvironment, invasion assays, mechanotransduction, STAT1/immune signaling, ECM remodeling

**Molecular & Analytical**
- RNA sequencing, qPCR, immunofluorescence, ELISA, western blotting, cytokine assays

**Quantitative & Imaging**
- Microfluidics, confocal microscopy, live-cell imaging, computational data integration

**Collaboration & Communication**
- Scientific writing, interdisciplinary teamwork, conference presentations, student mentoring

## LEADERSHIP & PROFESSIONAL DEVELOPMENT

- **COLMAN Inclusive Leadership Fellow,** Cornell University — *2024*
  Selected for a competitive fellowship program focused on building inclusive leadership skills and fostering diversity in STEM.
- Engaged in structured workshops and collaborative discussions aimed at enhancing team performance, equity, and belonging in scientific environments.

**WE Lead (Women in Engineering Leadership Program)**, Cornell University — *2023*
- Participated in a selective leadership program designed to advance women leaders in engineering and technical disciplines.
- Trained in strategic decision-making, negotiation, and leading complex technical teams.

**International Student Ambassador**, KTH Royal Institute of Technology — *2018–2019*

- Represented the Medical Biotechnology program globally; provided mentorship to prospective students on academics, research, and life in Sweden.

## SELECTED PUBLICATIONS & PRESENTATIONS

• Mechanical compression regulates tumor spheroid invasion into a 3D collagen matrix — Physical Biology, 2024.
• Enhanced extracellular matrix remodeling due to embedded spheroid fluidization — New Journal of Physics, 2025.
• Decoding physical principles of cell migration under controlled environment using

microfluidics — Biophysics Reviews, 2024.
• Viscoelastic properties of tissue spheroids revealed by a microfluidic rheometer — GRC Physical Science of Cancer, 2025.

## ONGOING RESEARCH

• STAT1 Signaling Under Mechanical Stress: Linking microenvironmental compression to immune signaling modulation in breast tumor organoids using RNAseq and immunostaining — exploring translational biomarker potential. (manuscript in preparation)

• Viscoelasticity in Tumor Organoids: Quantitative mechanical phenotyping to understand tumor stiffness as a predictor of invasive and therapeutic response behavior. (manuscript in preparation)